RESEARCH AND IMPLEMENTATION CONTRACT

Latent Causal Infeasibility for Long-Horizon TAMP

Methodology specification v1.6

Joint object selection, destinations and relocation order

The required outputs are an initial-task infeasibility prediction, an explicitly scoped intervention explanation, and ordered corrective actions. Robot joint trajectories are constructed and checked by the external motion backend. This is a revised design, not a claim that the new learned components have been implemented or validated.

Revision decision

This specification replaces v1.5 for new learned-model experiments and is paired with Implementation Guide v1.8. It keeps initial-scene inference without a reference target trajectory, ghost placements, full articulated checking, move-once repairs, unknown labels and the one-shot target-evaluation boundary. It adds explicit ordered repair variables, learned sequence scoring, required explanation outputs and a defined cost policy.

The current v1.5/v1.7 PDFs were read in full through their extracted text. G4A report 0008 and the subsequent conversation inform the migration. Repository source, raw G4A evidence and addendum A1 were not independently inspected for this revision; report statements are attributed historical facts. Preserve archived evidence and revalidate any reused label against its exact scope.

What must not be silently changed again

The model must represent the robot and action constraints, not merely a tidy endpoint arrangement. Ghosts substitute poses of existing objects. Object relevance depends on intervention and destination. Order is now part of a proposal, not an unreported executor choice. Learned scores are predictions; certificates and complete witnesses support stronger statements.

How to read this report

Sections 1–4 define the problem and physical evidence. Sections 5–9 define the network, explanation, energy, costs and solver. Sections 10–12 define supervision and deployment. Sections 13–16 give worked cases, evaluations, 3D scope and the precise migration.

The new protocol is `INITIAL_SCENE_ORDERED_ONE_SHOT_V2`. The previous `INITIAL_SCENE_ONE_SHOT_V1` and `FROZEN_CONTROL` remain separate historical protocols. No old result is relabelled as evidence of learned ordering or causal localization.

1. The problem, in ordinary language

The robot wants to place target T at a goal. A occupies that goal. B prevents access to A under the declared grasp family. C occupies a possible parking position P, while Q is free. A useful correction may be B to R, then A to Q. Choosing A to P instead may require moving C. This is an illustrative dependency, not a measured scene.

The three mandatory outputs

Initial infeasibility prediction. Estimate the prospects of completing the target action without corrective moves. The model returns a score and a validation-set operating decision, with an abstention option. Low predicted success is not an impossibility certificate.

Intervention explanation. Return object sets, the specified pose changes, their predicted effect on target feasibility, and the effect of omitting actions from a proposed repair. Each explanation names the baseline, intervention and scope. Distinguish original-task obstruction from participation in a corrective sequence. Necessary-cause and minimum-repair flags remain unasserted unless supported by sound evidence.

Ordered corrective actions. Return a finite sequence of object IDs and supported destination poses, with score, predicted motion cost and evidence status. The motion backend subsequently supplies grasps and joint trajectories. An unverified proposal is never presented as an executable witness.

Why these outputs interact

An initial diagnosis may suggest moving A, but A's destination determines whether C must also move. B may be relevant because it obstructs a corrective action, even if it does not occupy the target goal. Therefore do not freeze a culprit set before considering destinations and order. All physical objects, including those that stay, remain represented.

One-shot does not mean one hypothetical proposal

The system may score many hypothetical repairs from the initial observation. It freezes a ranked pool before external relocation checks. It commits the first complete relocation witness in that pool, executes it, and only then evaluates the target. A target failure ends this primary episode; it does not select another repair.

The learned order is an order proposal selected through learned energy. It is not a prediction of every joint waypoint, a causal DAG, or a guarantee of globally optimal planning.

2. Observations, object representations and robot state

Let S_0 be the complete resettable state and c_0 the learner-visible context. Use explicit field names for the task goal and gripper aperture.

$$c_0 = (I, \mathcal{M}, \{p_i, G_i, \text{support}_i\}, p_T, G_T, q_0, g_0, m_0, W, a, \text{goal}, C, \mathcal{D}).$$

Here I denotes RGB-D observations, $\mathcal{M}$ instance masks, p_i poses, G_i known geometry, q_0 robot joints, g_0 gripper gap, m_0 attachment state, W fixed geometry, a the target action, C constraints and $\mathcal{D}$ candidate poses. Include robot dimensions, limits and TCP/gripper geometry through a pinned numerical profile, not a learned hash embedding.

Object representation

A frozen visual backbone produces image features. Pool the features under each object's mask and apply a trainable projection to obtain z_i. A slot is an object-specific vector; using supplied masks is not Slot Attention. Slot Attention is a later optional perception replacement requiring its own instance-association and occlusion evaluation [1]. The core study assumes supplied primitive geometry and poses, with provenance.

A missing/fully occluded mask is a missing-data condition. Use an explicit missingness indicator or a declared eligibility rule; never interpret an all-zero vector as observed appearance. Keep stationary objects and nonactive targets in collision context. The active target is not a repair object in this benchmark.

Robot and workspace representation

For the fixed suite, encode normalized joint angles, joint limits, link/gripper dimensions, attachment and gap, and numerically derived forward-kinematic link poses. Encode fixed geometry by declared primitive descriptors or fixed, geometry-only samples. These are inputs, not collision-oracle results. Changing the profile requires a new context and hash.

Forbidden shortcuts

No target trajectory, swept envelope, oracle distance, collision label, teacher-selected destination, root/family ID, planner seed, construction ancestry, future image or achieved teacher joint state enters proposal tensors. Object IDs associate records but have no numerical/ordinal meaning. Clean images exclude paths, labels and diagnostic overlays.

The proposal uses initial observations only. Hypothetical prefix scenes are deterministic substitutions from its own proposed actions; they are not fresh observations or privileged simulated outcomes.

3. Ghost placements and canonical ordered repairs

For each movable obstacle i, the finite domain is $\mathcal{D}_i = \{0, 1, \ldots, K_i\}$: zero means its actual initial pose and $k > 0$ refers to a supported physical pose. Deduplicate pose-equivalent entries and exclude a duplicate of stay. Per-object indexes are not global cell IDs. Do not reject an occupied destination solely because its occupant may move in the same repair.

A ghost is a hypothetical pose of the same object: the old occupancy is removed and the proposed occupancy inserted. All candidates are alternative values, never simultaneous clones. Changing a destination, restoring an object to stay, and deleting an object are distinct operations. Physical deletion is not a legal primary repair.

Use one canonical sequence, without padding duplicates

Represent a proposal as

$$r = ((i_1, k_1), \ldots, (i_m, k_m)), \qquad 0 \le m \le b,$$

where IDs i_t are distinct and $k_t > 0$. The empty sequence $\varnothing$ means no repair. The final assignment $x(r)$ is derived: selected objects use their chosen pose and all others stay. The sequence itself defines the order. Do not maintain a second independent permutation that can disagree with it.

This representation avoids the multiplicity introduced by padding many equivalent NOOP layouts or permuting inactive objects. Padding is allowed inside minibatches only, with a length mask; it is never a distinct sampler state. A physical signature is the ordered list of stable object identities and physical poses. Swapped orders remain distinct even if their endpoints agree.

What an action means

Each pair (i_t, k_t) requests a complete relocation skill: pregrasp, approach, closure/attachment, supported carry, release/detachment and safe retreat. It does not specify grasps or joint paths. The executor must preserve this order. Another order can be another ranked proposal, but cannot be silently substituted inside verification of r.

The ideal prefix scene places previously selected objects at their proposed destinations and leaves other objects at initial poses. It helps the learned scorer understand access changes. It does not contain a known intermediate robot configuration and is not an execution witness.

One-hot candidate indicators and a categorical destination variable are equivalent encodings. Neither is an extra physical object. No hidden/auxiliary BM-unit reduction is assumed in this design.

4. Physical feasibility, scope and complete witnesses

Static endpoint validity checks support, fixed workspace, the resting target and simultaneous obstacle endpoints. It does not check that a motion exists. An endpoint may occupy a region initially occupied by the arm if the arm can move away; do not freeze the initial robot geometry as a permanent obstacle.

For an ordered repair r, let $\mathcal{W}(r; S_0, C)$ contain all allowed continuous relocation realizations with exactly that object order and those destination tolerances. A realization w includes grasps, joint paths, attachments and achieved states. Define

$$Y^*(r; S_0) = \mathbf{1}[\exists w \in \mathcal{W}(r; S_0, C),\ \exists \gamma : \text{Valid}(w, \gamma) = 1],$$

where γ is a complete target action from the achieved final state of w. The empty-repair label is original target feasibility. The target has no prescribed reference trajectory.

Evidence is three-valued

Positive: a complete independently checked w and target γ. **Negative:** a sound certificate excludes every allowed continuous realization of this particular r. **Unknown:** neither is established. A collision in one trajectory, a failed IK branch, a blocked connector from one intermediate joint state or a timeout does not exclude all realizations of an ordered proposal.

An assignment-wide certificate valid for every order can label all corresponding ordered proposals negative. A witness for one order labels only that order positive. A fixed-path negative labels that path, not automatically r. Keep these scopes distinct in code and training.

Complete articulated checks

Preserve the tested 3R profile and reported 48-candidate grasp family; read actual values from the repository. Check links, housings, base, palm, fingers, attached object, self-collision and fixed/stationary bodies. Carry forward achieved joints, gap and attachment at every transition; no free reset to home. Use float64 geometry and apply clearance once.

Goal achievement plus checked safe retreat is the new target completion policy adopted from the reported A1 contract. HOME return is a separate diagnostic. Hash the policy; HOME-required historical labels remain separate. Continuous verification requires analytic/conservative checks, not merely collision-free sampled nodes. Unresolved numerical intervals remain unknown.

Kinematic witnesses, dynamic/controller simulation and hardware success are separate evidence levels. This revision authorizes no hardware actuation.

5. A minimal robot-aware learned scorer

The model learns scoring functions; the solver searches the resulting energy. No target planner is embedded in a neural forward pass.

Static branches

Construct $h_i(k; c_0)$ from z_i, shape, initial and candidate poses, displacement, support, task-relative geometry and robot context. Shared MLPs predict unary support/workspace logits and pairwise endpoint compatibility logits. Enforce object-exchange symmetry in the pair branch. All stay factors remain present.

Ordered branch

For each action t, pool representations of *all* objects in the ideal prefix scene, including stay objects. Concatenate this scene summary, the selected object/candidate token, numerical robot/action context and step position. Process these action tokens in order with a single width-128 GRU. Its initial hidden state is a projection of the initial scene and robot/task context. Use a final scene pool and sequence length in the readout.

The two readouts are a complete ordered-repair logit $u_\theta(r, c_0)$ and a nonnegative predicted relocation-motion cost $\widehat{J}_\theta(r, c_0)$. The empty sequence uses the same initial/final-scene readout; its relocation cost is exactly zero.

$$\widehat{p}(r \mid c_0) = \mathrm{sigmoid}(u_\theta(r, c_0)), \qquad e_{\mathrm{task}}(r, c_0) = \mathrm{softplus}(-u_\theta(r, c_0)).$$

What this does and does not represent

The recurrent state is a learned summary of a proposed action prefix. It is **not** an asserted intermediate joint configuration, a dynamics rollout or a collision certificate. Order-specific labels and initial robot information allow the network to learn execution prospects. The backend constructs the actual joint sequence and validates continuity.

Object-record order must not matter: consistent slot permutations leave scores unchanged. Action order must matter: changing B-then-A to A-then-B may change the score. These are different invariances and require separate tests.

The GRU is an implementable starting choice, not an expressivity claim or proven optimum. If ordered labels are too sparse to teach it, stop at the data gate rather than replacing unknowns with failed-search negatives. No separate continuous destination decoder or joint-trajectory decoder is included.

6. Explanation is an explicit output, not a weight interpretation

Use the same task scorer for original feasibility and controlled counterfactual scenes. The initial prediction is $\widehat{p}_0 = \widehat{p}(\varnothing \mid c_0)$; the infeasibility score is $1 - \widehat{p}_0$. Report its calibration and abstention policy. Do not call a sigmoid output a certificate.

A. Scene-intervention effect

Let I specify one or more object pose substitutions. Construct c_0^I with those poses while holding robot joints, gap, attachment, task and other scene variables fixed. The hypothetical scene must be well-defined: supported, nonpenetrating and compatible with the fixed initial robot state. Otherwise the comparison is ineligible, not an automatically negative task example.

$$\widehat{\Delta}_{\text{scene}}(I) = \widehat{p}(\varnothing \mid c_0^I) - \widehat{p}(\varnothing \mid c_0).$$

This predicts the effect of a stated pose intervention on target feasibility *at fixed robot state*. It does not prove the robot can perform that intervention. Never substitute a successful repair's final joint state into only one side and call it an object-only effect.

Cache appearance vectors from the original image/masks for both sides; change structured poses and recompute geometric context only. Apply this same anchored-appearance rule during training. A rendered counterfactual image is diagnostic evidence, not an extra deployment observation. This is a learned geometric counterfactual conditioned on observed appearance, not a claim of image synthesis.

B. Omission effect within an ordered repair

Delete action t from r, preserve relative order of the others and restore that object's initial pose in every recomputed prefix. Define

$$\widehat{\Delta}_{\text{omit}}(t \mid r) = \widehat{p}(r \mid c_0) - \widehat{p}(r \setminus t \mid c_0).$$

This predicts the importance of that action for *this proposed repair*. It is not universal object necessity. Include joint omissions when individual effects miss interactions; log all tested subsets and the finite explanation budget.

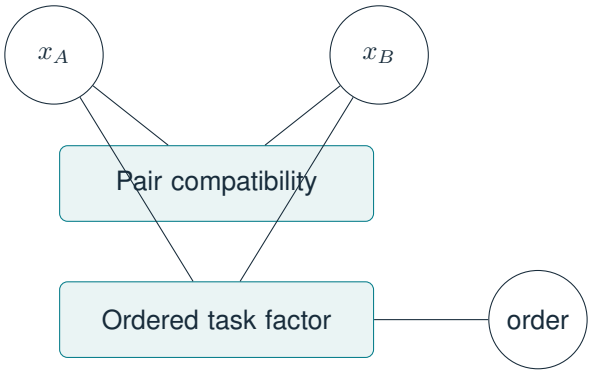
Required explanation record

For each returned entry store object IDs, original/proposed poses, intervention type, conditioning scene/repair, baseline and alternative scores, signed effect, eligibility and evidence status. Predict singleton scene effects where valid, the full selected joint intervention, single-action omissions and selected pair omissions. Keep negative effects as well as positive ones.

The learned output is a scoped effect estimate. Optional independent certificates can attach statements such as mandatory goal occupancy, grasp exclusion or a minimum bound. Do not invent a semantic role from attention, energy magnitude or move frequency. A necessary-cause field defaults to UNESTABLISHED. Explanations do not permanently prune objects or destinations.

7. The factor graph and the energy equation

The factor graph is the dependence structure of one energy model. Variable nodes describe hypothetical decisions. Unary factors inspect one endpoint, pair factors inspect two, and the ordered task factor inspects the whole sequence. This is not a causal DAG. A higher-order factor can be drawn as a hyperedge; no auxiliary binary reduction is required.



The diagram shows derived destination/order decisions for readability. The implementation stores one canonical sequence r. Unary factors and other objects are omitted only from the drawing.

$$E_{\text{feas},\theta}(r \mid c_0) = \sum_i e_i^w(x_i(r)) + \sum_{i<j} e_{ij}^p(x_i(r), x_j(r)) + \lambda_{\text{task}} e_{\text{task}}(r, c_0).$$

For a predicted compatibility $q = \text{sigmoid}(\ell)$, $e = -\log q = \text{softplus}(-\ell)$. Adding penalties is equivalent to multiplying positive factor scores before taking a negative logarithm. This constructs a useful compatibility model; correlated local/global factors do not constitute an exact factorization of physical success probability.

The model designer declares the factors and constraints; supervision learns their functions. At deployment weights θ are fixed and decisions r change. Hopfield and Boltzmann approaches are related energy-based methods, but this higher-order neural factor model uses stochastic Boltzmann inference, not a separately implemented classical Hopfield network [2,3].

Local updates and global evaluation use the *same* energy. A destination change requires all affected pair terms and the whole ordered task term to be recomputed. A sequence update can affect every subsequent prefix; the old final-sum-only cache is insufficient.

8. Cost: few moves, short displacements and robot motion

Feasibility is not bought with a cheap distance

Let $m(r)$ be the number of distinct moved objects. Keep the declared move cap b as a hard search-domain constraint. Exact static rejection remains outside learned certainty. Among candidates passing the frozen task-score threshold and static checks, the primary ranking is lexicographic: fewer moves, lower predicted robot-motion cost, shorter endpoint displacement, lower feasibility energy, then a deterministic physical-signature tie break.

Known endpoint displacement

With workspace diagonal $L_W > 0$, define

$$C_d(r) = \sum_{(i,k)\in r} \|p_i^k - p_i^0\|_2 / L_W.$$

It is dimensionless, geometry-computed and does not need learning. In 3D use the declared translation coordinates; yaw distance is a separately reported quantity unless explicitly included in a new cost policy. This is not carried-path length or robot effort.

Measured robot-motion cost and learned prediction

For a checked relocation witness with continuous joint-coordinate waypoint increments, use

$$J(w) = \sum_{\text{segments } s} \sum_{j} |q_{s+1,j} - q_{s,j}| / (q_j^{\max} - q_j^{\min}).$$

Every configured joint range must be finite and nonzero. Use the backend's actual unwrapped joint coordinates and interpolation; no fictitious jump across angle wrapping. Include approach, carry, release-associated arm motion, retreat and transitions for all relocations. Gripper aperture, elapsed time and the subsequent target motion are separate diagnostics. This is normalized joint travel, not energy consumption or torque.

The cost head predicts the normalized travel of the declared finite relocation backend's first complete witness for r, conditional on finding one. Pin seed schedules and backend policy; do not select the lowest-cost or target-successful witness and silently change this target. Unknown/missing relocation witnesses have no cost label. Predicting cost does not certify feasibility or minimum possible path cost.

Sampling score

$$E_{\text{sample}} = E_{\text{feas}} + \lambda_d C_d + \lambda_q \widehat{J}.$$

Use explicit nonnegative, validation-frozen coefficients. Default starts are $\lambda_d = 0.1$, $\lambda_q = 0.1$, $\lambda_{\text{task}} = 1$. These are proposed settings, not results. Finite-pool feasibility training uses E_{feas}, not the cost-augmented score. Final lexicographic selection preserves the primary move-count priority; sampling does not guarantee its optimum.

9. Joint Boltzmann search, including order

On the finite canonical sequence space $\mathcal{R}_b$, define

$$P_{\theta,b}(r \mid c_0) = Z^{-1}\mathbf{1}[r \in \mathcal{R}_b] \exp[-E_{\text{sample}}(r \mid c_0)/T], \quad T > 0.$$

Weights remain fixed. Search changes hypothetical sequences, never physical state. Use four fixed-probability kernels (initially $1/4$ each); an unavailable move is a self-transition. Avoid hard static masks inside the chain initially: screening at collection preserves passage through temporary incompatible hypotheses.

Destination Gibbs. Choose a selected action uniformly and resample its destination over that object's K_i nonstay values, holding order and other actions fixed. Compute the full score for each value. This is a conditional update of the joint distribution, not independent optimization.

Order swap. Choose two distinct action positions uniformly and swap them. The proposal is symmetric; accept with $\min(1, \exp[-(E' - E)/T])$. Recompute affected prefixes and the ordered score. This can change execution prospects despite identical endpoints.

Birth. When $m < b$, choose an unused object uniformly, one of its K_i destinations uniformly, and one of $m + 1$ insertion positions uniformly. **Death.** Choose one of m actions uniformly and delete it. Neither operation deletes an object from the scene; death restores stay.

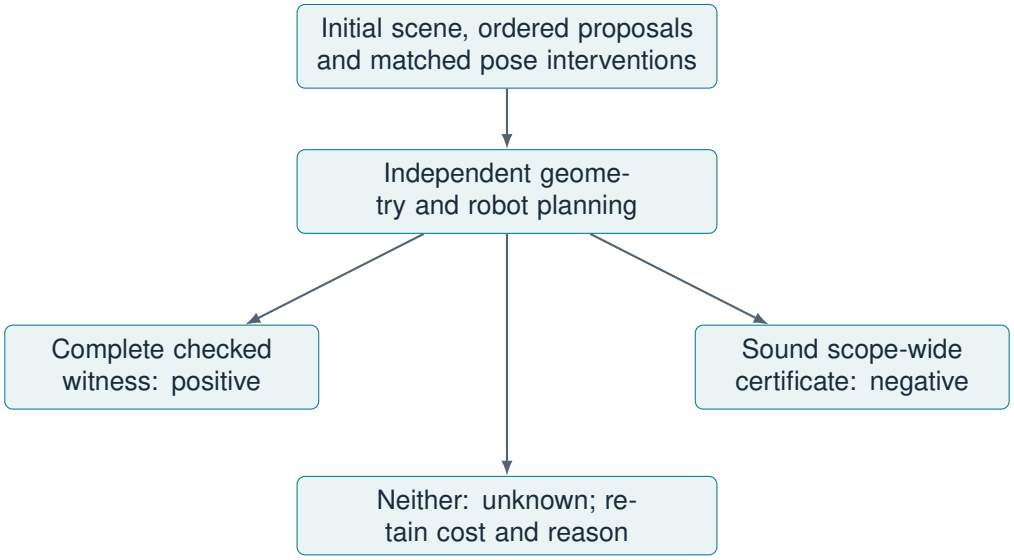
These last proposals are asymmetric. Use Metropolis–Hastings:

$$\alpha(r, r') = \min\left(1, e^{-(E'-E)/T} \frac{Q(r \mid r')}{Q(r' \mid r)}\right).$$

For a birth of object i from length m, with equal fixed birth/death selection probabilities, the reverse/forward ratio is $(N - m)K_i$. For a death of i from length m, it is $1/[(N - m + 1)K_i]$. Here N counts eligible objects with at least one nonstay candidate; other physical objects remain context. Compute ratios in log space and test variable-K_i cases. If proposal rules change, rederive Q.

Birth/death connect different move sets; swaps explore order; Gibbs explores destinations. At fixed positive temperature and finite soft scores the intended finite-state chain can traverse these states, but a bounded run is not guaranteed mixed, optimal or complete. Deduplicate collected sequences physically, include empty and mixed initializations, and freeze a finite pool/ranking before external checks. G4B must verify transition probabilities against tiny exact enumeration, not merely show a pleasant sample animation.

10. Offline evidence and causal comparisons



The teacher may search target outcomes offline. It must restart every proposal from the same recorded initial state, preserve the prescribed order, and plan each next action from the previous achieved joint/object state. A completed relocation witness contributes a motion-cost label even if target feasibility remains unresolved; the label refers to that declared first-witness policy.

Generate target-only labels for matched, valid counterfactual scene edits at fixed robot state. These supervise the empty-sequence scorer on both sides of the scene-effect equation. An effect label $\Delta \in \{-1, 0, 1\}$ is definite only when both binary outcomes are decided. If either side is unknown, mask the effect loss and retain the pair for coverage reporting.

Generate paired ordered sequences with the same endpoints but different orders, and repair-omission comparisons. A failed sampled trajectory cannot be used as a negative for the entire order. Useful order negatives require sound first-action access/placement exclusion or another proof covering all allowed realizations for that order. If such decided contrast pairs cannot be obtained, report an order-supervision gate failure.

Retain multiple successful alternatives rather than one teacher-chosen culprit mask. Include joint interventions, no-change edits, nuisance edits and alternative destinations. Group every scene descendant, rendering, order, cost sample and intervention pair under its root split. Splitting after data mining leaks related scenes across train and test.

11. Training objectives and what learns what

Static factors learn from decided geometric labels. The sequence head learns from ordered complete-repair evidence, including empty repairs and diagnostic scene substitutions. The cost head learns from complete relocation witnesses. No gradient passes through IK or the geometric checker.

$$\begin{aligned}
\mathcal{L}_{\text{static}} &= \text{mean}_{f \text{ decided}}\, \text{BCEWithLogits}(\ell_f, y_f),\\
\mathcal{L}_{\text{task}} &= \text{mean}_{r \text{ decided}}\, \text{BCEWithLogits}(u_\theta(r), Y^*(r)),\\
\mathcal{L}_{\text{effect}} &= \text{mean}_{\text{decided pairs}}(\widehat{\Delta} - \Delta)^2,\\
\mathcal{L}_{\text{cost}} &= \text{mean}_{\text{relocation witnesses}}\, \text{Huber}(\log(1+\widehat{J}), \log(1+J)).
\end{aligned}$$

The effect loss applies to both valid fixed-robot scene pairs and valid repair-omission pairs, with their scopes kept separate. The same shared scorer produces the compared values. Thus explanation is an explicit learned and evaluated function, without inventing a causal probability from network weights.

For each deduplicated mixed pool C_{dec} of decided ordered proposals from the same initial query, let C_+ be all positives. Use

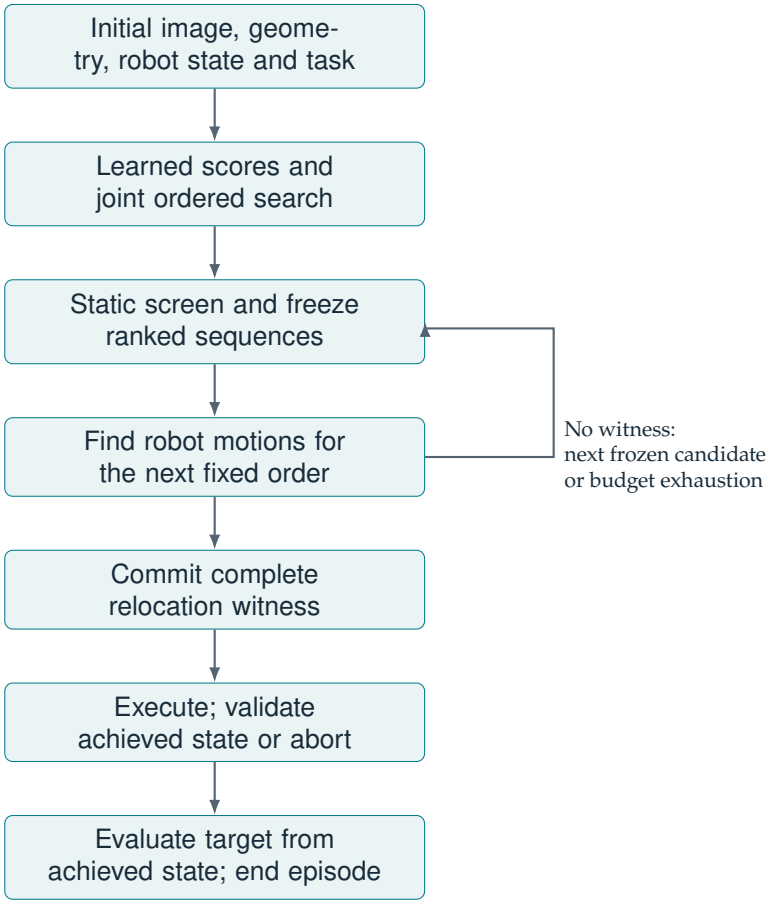
$$\widetilde{P}(r) = \frac{e^{-E_{\text{feas}}(r)/T_{\text{train}}}}{\sum_{v \in C_{\text{dec}}} e^{-E_{\text{feas}}(v)/T_{\text{train}}}}, \quad q_C(r) = \frac{\mathbf{1}[r \in C_+]}{|C_+|}, \quad \mathcal{L}_J = -\sum_r q_C(r) \log \widetilde{P}(r).$$

Unknowns are excluded from the joint denominator as well as targets. Skip the joint term for all-positive/all-negative pools. Its independent-energy derivative is $(q_C - \widetilde{P})/T_{\text{train}}$; test the sign numerically. Uniform-positive mass teaches feasibility, not minimum cost; cost learning and selection remain separate.

$$\mathcal{L} = \mathcal{L}_{\text{static}} + \lambda_s \mathcal{L}_{\text{task}} + \lambda_\Delta \mathcal{L}_{\text{effect}} + \lambda_c \mathcal{L}_{\text{cost}} + \lambda_J \mathcal{L}_J.$$

Warm up static/task/effect/cost heads, then introduce joint loss. Initial loss coefficients are all 1; tune only on validation and log actual supervision counts. Empty label groups contribute zero with an explicit count, never fabricated targets. Use root-balanced batches and retain hard dependencies and multiple positives. Freeze the visual backbone initially. This is supervised neural energy learning, not classical contrastive-divergence training of a BM.

12. Deployment and the no-target-feedback boundary



The explanation outputs are computed from learned counterfactual comparisons using initial-context substitutions only. No online target calls are used to validate them before selection. Any explanatory ground-truth audit happens independently after the primary decision is fixed.

Among static-valid candidates with $\widehat{p} \geq \tau$, freeze the ranking from Section 8. Threshold τ, abstention rule and computation budgets are chosen on validation and frozen before test. No candidate passing the threshold means ABSTAIN, retained in the denominator. Empty repair needs a favorable task score; otherwise zero moves would always win.

The relocation verifier may vary grasps and paths within the specified order, but may not reorder it. Before physical execution it may advance through the frozen pool on relocation failure without target feedback. After commitment, selections, destinations and order stay fixed. Low-level replanning to the same next action is separately logged and budgeted.

Each target invocation is a complete predeclared finite evaluator schedule, not one IK attempt. There are zero primary target invocations before repair completion, and at most one afterward. Abort/no-witness episodes have none. Target failure does not restart the repair. HOME diagnostics are separately identified and cannot alter success or selection.

13. Worked examples and the limits of each conclusion

A. Goal occupancy plus access dependency

Suppose a unique required goal overlaps A for every allowed final target pose. An independent goal certificate establishes that A must move. Suppose every permitted first grasp of A is excluded while B stays. That can exclude an A-first ordered proposal when the proof covers its full grasp/contact domain. A checked B-to-R, A-to-Q sequence plus the target establishes sufficiency. It establishes a two-move optimum only if an appropriate lower bound also excludes every one-move repair.

The learned outputs may predict low no-repair feasibility, high effect for the joint intervention, and a strong omission effect for B in this repair. Those predictions do not themselves establish the certificates. B's omission effect is not automatically a direct-target obstruction label.

B. Destination choice changes relevance

C occupies P. A-to-P with C staying is endpoint-invalid. A-to-Q may allow C to stay. Thus C is relevant to one chosen destination but not necessarily to the task's minimum repair. A single teacher mask labelling C as always causal would be incorrect. Keep both successful alternatives if witnessed; compare move counts and declared costs.

C. A successful pose edit is not an executable correction

Teleport A to Q while holding the robot at q_0, and suppose the target becomes feasible. This supports a diagnostic scene-effect label if the original side is decided. It does not prove that the robot can relocate A, nor that the robot state left by that relocation admits the same target motion. A complete ordered witness is a separate requirement.

D. Endpoint distance and robot cost disagree

Q is closer to A than R, but reaching Q may require a longer arm detour or difficult retreat. The endpoint-distance term favors Q. The motion-cost predictor may favor R after training on checked relocation traces. Neither can overrule exact validity. Report the chosen tradeoff and achieved joint travel rather than calling endpoint distance robot effort.

E. Different continuous realizations of the same order

One B-then-A execution leaves a poor final joint configuration for T; another may permit T. A witness proves $Y^*(r) = 1$, but the online finite executor may find a different realization and fail afterward. This gap is retained in the operational metric. Ordered variables reduce ambiguity; they do not optimize every grasp or final joint configuration. Do not hide the gap with target look-ahead.

14. Evaluation, baselines and justified claims

Preserve B0–B9, O1 and F0 historical definitions. Record the old method/protocol in comparisons; an endpoint method with external order search is not the same learned system as V2. Add explicitly named V2 controls rather than silently reusing result labels.

Minimum comparisons are no repair; target-blind geometry proposals; uniform ordered sequences; learned endpoint proposals with external order search; the full ordered model; the full model with fixed/random order under matched proposal and motion budgets. Include a retrained order-insensitive model to isolate learned order rather than only corrupting a trained model at inference.

For claims about components, compare geometry-only versus mask features, no pair factor, no ordered task factor, no effect loss, no cost head and no joint loss. Removing a component at inference and retraining without it answer different questions. Keep root, label coverage, candidate domains, planner/backend and budgets matched. Privileged oracle scoring is O1, never an equal-information primary baseline.

Measure each required output

Infeasibility: performance and reliability on decided no-repair cases; report unknown coverage and abstention, not timeout-as-negative accuracy. **Explanation:** error/sign on decided paired effects, separate singleton/joint and scene/omission scopes, predicted versus checked sufficient sets, and certificate-backed necessity only where available. No teacher-set overlap presented as universal causal accuracy.

Repair: full target success over every eligible episode, including abstention, abort and unknown; unnecessary moves on independently witnessed initially feasible cases; relocation success; move count; endpoint distance; measured joint travel; proposal, scoring, planning and execution costs. Order ranking is evaluated on same-endpoint alternatives with decided evidence and coverage reported.

Cost prediction is evaluated on successful relocation traces under the same backend; show error by move count, object density and robot state. Best-found cost is not a proven motion optimum. Match planning effort when measuring cost improvement.

Use independent-root splits and root-cluster confidence intervals. Dense sibling variants are not independent roots. Keep test outcomes out of thresholds, certificate mining and model choice. Report geometric-baseline wins honestly. G4A's 8/20 geometric result is neither a learned baseline result nor an estimate of V2 performance.

15. 3D, long horizon and unchanged physical guardrails

2D contract

Use the actual validated fixed-base 3R profile, full finite-width arm and gripper, reported 48 grasps, actual tested home and tolerances. Retain the 8 by 8 endpoint grid, 0.08 m pitch, 0.024 m obstacle radius and 0.005 m margin unless a separately versioned experiment changes them. Distinct grid cells imply pair clearance only for this geometry. Motion between cells remains continuous. Sparse cases precede 30–40-object density challenges.

3D cupboard contract

Pin complete xArm7/gripper assets, TCP, mount, joint limits, collision exclusions and contact policy. Proposed cupboard dimensions remain 0.64 m square, support top 0.005 m and roof underside 0.30 m; validate them against actual assets. Upright cylinders have radius 0.024 m and height 0.075 m, giving supported center height 0.0425 m. Initial proposed tolerances are 0.001 m height and 0.035 rad tilt, to be validated before use.

Supported attached transport forbids vertical overflight. The robot links themselves move in 3D and are not constrained to the support plane. Check walls, roof, support, arm, gripper and attached object throughout motion. Start with 6–10 obstacles and sparse equivalents of goal, access, destination and order cases. Keep controller dynamics and kinematic replay separate. Simulation attachment is not proof of a physical grasp.

Train and evaluate 3D separately first. A 2D-to-3D transfer experiment is additional, not assumed. No 2D certificate automatically transfers to the 3D contact model. Hardware requires separately authorized calibration, uncertainty/contact assessment and trials.

Long horizon

G10 carries achieved states through at least three original task actions. Each new target episode obtains a new initial observation; within one repair episode its high-level proposal remains fixed. Move-once history resets only on completion or explicit abandonment. A repair may help the current goal and obstruct a later one. There is no future-plan-suffix energy in this revision; measure that limitation rather than claiming global long-horizon optimality.

No online RL, unrestricted continuous destination decoder, pushing, repeated relocation of one object, learned contact dynamics, fluid simulation or articulated-door benchmark is silently added. Such actions require their own model and evidence contracts.

16. Migration, readiness and sources

Changes from methodology v1.5

The output changes from an endpoint assignment to a canonical ordered relocation sequence plus initial infeasibility and explicit intervention-effect explanations. The sum-pooled assignment task head becomes a permutation-invariant scene encoder followed by an order-sensitive GRU. Initial robot state remains an input; continuous intermediate robot states remain backend evidence, not guessed neural facts.

Destination Gibbs is retained for fixed selected actions, while order swaps and birth/death Metropolis–Hastings moves search order and move-set size. The proposal density includes the correct asymmetric proposal ratio. Endpoint-only Gibbs claims and cache assumptions no longer apply unchanged.

Costs are now explicit: primary move count, known normalized endpoint distance and predicted normalized relocation joint travel. Feasibility training and cost learning have distinct targets. A frozen ranking and no-target-feedback execution remain mandatory. Target success requires goal achievement plus safe retreat, with HOME separate.

What remains valid from G0–G4A

Scene, robot, geometric checker, evidence-audit and protocol-boundary code can be reused after inspection. Report 0008 records G4A as NOT_READY_TO_SCALE, with 44/86/110 assignment labels and three zero-move roots against six. These are reported V1 outcomes. The separate three-root teacher-budget follow-up remains bounded and does not authorize V2 scale or training.

Create a new G4D design/data migration checkpoint before V2 sampler or training work. Do not overwrite archives. An old positive can supervise its recorded order only after validating exact state, policy and complete witness. An assignment-wide certificate may transfer to orders only if its premises and scope match. Old unknowns remain unknown. Existing labels do not automatically supply the new diagnostic scene pairs or first-witness cost targets.

Research status

This document fixes a testable architecture, not evidence that causal localization, useful ordering or cost reduction will succeed. Data readiness, finite-sampler correctness, matched ablations and operational evaluation determine those claims. The companion guide specifies the gates and stop rules.

Sources and terminology

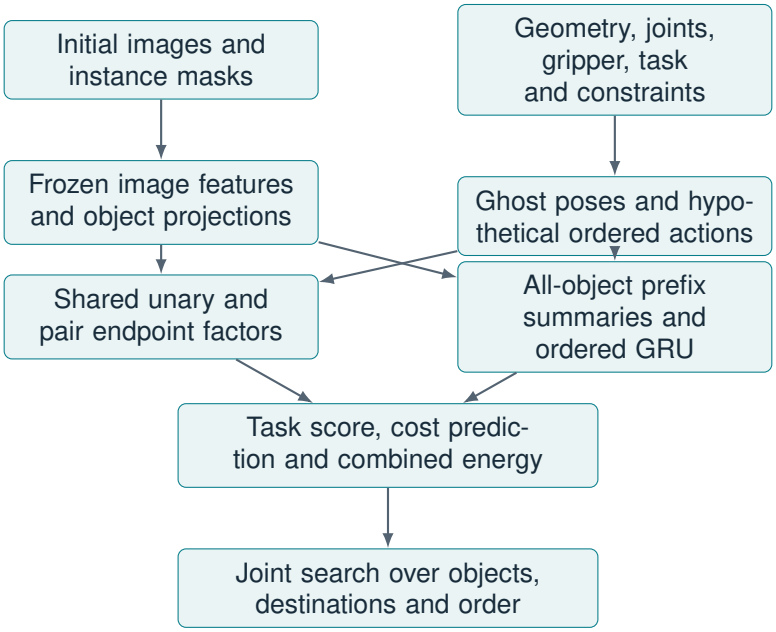
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[2] Hinton, *Boltzmann Machines*, 2007. https://www.cs.toronto.edu/~hinton/csc321/readings/boltz321.pdf.

[3] Hopfield and Tank, *Neural computation of decisions in optimization problems*, 1985. DOI: 10.1007/BF00339943.

Project sources: methodology v1.5, implementation guide v1.7 (16 October 2026); report 0008; the G4A follow-up instruction; and the user's clarification that infeasibility, explanation and executable corrective actions are core outputs. The older master is historical motivation only.

Appendix A. Read the complete learned architecture



Top row: what is observed. The masks separate the image information belonging to each object. The structured state says where things are and what the robot can do. The model does not infer known joint angles from pixels.

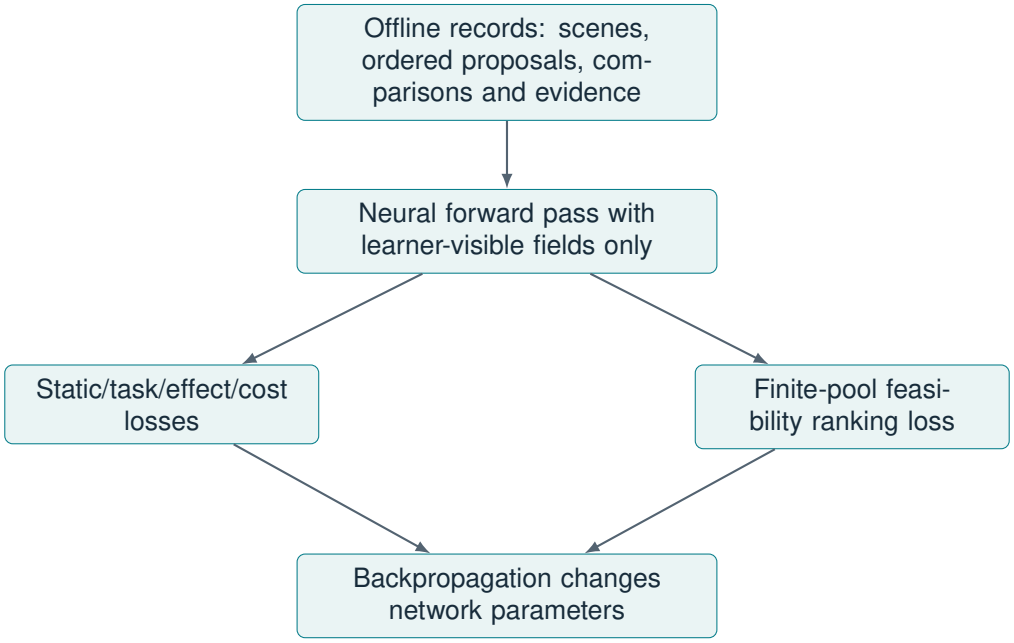
Middle row: what is imagined. A ghost is an alternative pose. An ordered candidate imagines which object moves first and where each selected object ends. Prefix summaries update hypothetical object geometry; they do not pretend to know the future arm joints.

Scoring row: what is learned. Local factors learn endpoint compatibility. The GRU sees the proposed action order and predicts complete execution prospects and relocation cost. These functions share object information but answer different questions.

Search row: what is solved. Boltzmann inference changes the proposed sequence while weights stay fixed. It feeds new candidates back to the same scoring branches. The endpoint and order choices are coupled through the energy, not solved independently and concatenated afterward.

The initial infeasibility output is the empty-sequence score. Explanation outputs come from additional specified scene/omission comparisons through the same scorer. The returned ordered sequence still needs the external checks in Section 12. This drawing is a design diagram, not evidence of a trained system.

Appendix B. Read the training and intervention loops



The teacher supplies evidence, not the answer as an input

Offline planning can evaluate B-then-A, A-then-B, different destinations and no repair. Complete traces supply labels and cost. The loader strips those outcomes and trajectories from the model inputs. The neural networks learn to predict them from the initial context and a hypothetical proposal.

A paired explanation example

For a scene-effect pair, compare the original scene with a specified pose substitution while holding the robot fixed. For an omission pair, compare B-to-R then A-to-Q with A-to-Q alone, leaving B at its initial pose. These are different experiments with different labels. Both sides must be decided before claiming a definite effect. Retain joint comparisons when single edits leave the target blocked.

What each loop changes

Training changes weights. Sampling changes hypothetical action sequences. Execution changes the robot and physical object states. Diagnostic pose interventions change a cloned scene for a scoped comparison. Confusing these loops would turn a predicted score into an alleged certificate, an imagined ghost into a physical move, or a teacher trace into leaked deployment information.

How order is learned

The same endpoints appear with different orders and independently checked outcomes. The recurrent scorer can therefore learn that an earlier action creates access for a later one. It does not get a target-planner answer while choosing the deployment repair. If the dataset does not contain decided order contrasts, the claimed learned dependency is not established.

A falling training loss, a frequently sampled object, a low energy and a complete physical witness are four different observations. Report each at its actual meaning and scope.